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Prediction of π-turns in proteins using PSI-BLAST profiles and secondary structure information

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Abstract

Due to the structural and functional importance of tight turns, some methods have been proposed to predict γ-turns, β-turns, and α-turns in proteins. In the past, studies of π-turns were made, but not a single prediction approach has been developed so far. It will be useful to develop a method for identifying π-turns in a protein sequence. In this paper, the support vector machine (SVM) method has been introduced to predict π-turns from the amino acid sequence. The training and testing of this approach is performed with a newly collected data set of 640 non-homologous protein chains containing 1931 π-turns. Different sequence encoding schemes have been explored in order to investigate their effects on the prediction performance. With multiple sequence alignment and predicted secondary structure, the final SVM model yields a Matthews correlation coefficient (MCC) of 0.556 by a 7-fold cross-validation. A web server implementing the prediction method is available at the following URL: http://210.42.106.80/piturn/.

Keywords: π-Turns; Support vector machine; Multiple sequence alignment

The gap between the number of known sequences and the number of known structures is widening because of the progress of large-scale genome sequencing projects. Developing accurate methods for protein structure prediction is necessary and important to bridge the gap. Protein secondary structure prediction is a key step in many different approaches to predict tertiary structure of proteins, especially in the case when the sequence similarity is lower than 30%. The protein secondary structure consists of regular elements such as α-helices and β-sheets, and irregular elements such as β-bulges, random coils, and tight turns. Tight turns are generally classified as δ-turns, γ-turns, β-turns, α-turns, and π-turns according to the number of residues involved [1]. They can help a polypeptide chain to fold into a compact globular structure by serving as a linker between different secondary structures, such as α-helices and/or β-sheets [1]. Furthermore, due to their frequent occurrence on the protein surface, they are usually relevant with molecular recognition. Because of the structural and functional importance of tight turns, a few approaches have been proposed to predict γ-turns [2], β-turns [3–9], and α-turns [10–13] in proteins. To our best knowledge, there has been not an approach to predict π-turns up to now. The present study provides, for the first time, a robust method to predict π-turns from a given amino acid sequence.

π-Turn corresponds to a chain reversal involving six amino acids and may be stabilized by a hydrogen bond between the CO group of the first residue and the NH group of the sixth residue [14]. It was described as a helix terminator with residue five in the α_L conformation before 1996 [15–18]. Rajashankar and Ramakumar [14] conducted a detailed analysis of 486 π-turns in 141 proteins. They found that the fifth residue of π-turns can adopt not only the α_L conformation but also α_R and β conformation. They also found the occurrence of the π-turn is not limited to the helix termini. A small number of π-turns were also noticed to occur at the ends of β-strands.

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In this study, we build an SVM-based classifier to identify π-turns with high accuracy. The SVM method has been successfully applied to many biological problems, such as protein secondary structure prediction [19,20], protein subcellular location prediction [21–23], protein solvent accessibility prediction [24,25], and enzyme classification [26,27]. The present SVM approach together with PSI-BLAST position-specific matrices [28], and secondary structure information predicted by PSIPRED [29] gave an MCC of 0.556. In addition, we computed the updated positional potentials of 20 amino acids of π-turns in the newly collected 640 non-homologous protein chains.

Materials and methods

Data set. The representative protein data set was selected from the pdbselect [30] of October 2005 with a sequence homology value <25%. Protein chains determined by X-ray crystallography were further filtered with conditions such as resolution better than 2.0 Å, refined with an R factor $\leqslant 0.23$, and containing at least one π-turn. After these filtering procedures, we obtained the final data set of 640 proteins including 1931 π-turns and 124,352 residues. The total number of amino acid residues consisting of π-turns in our data set is 9.3%. On average, there are three π-turns in each protein chain. The PDB codes corresponding to these representative protein chains are given in the supplementary material.

We identified π-turns with the following rules. The polypeptides containing six residues and having a hydrogen bond between residue i and residue $i+5$ were selected. Furthermore, the consecutive π-turns (π-helix) were excluded.

Calculation of position-specific propensities of 20 amino acids for π-turns. The amino acid propensities of π-turns were calculated using the following formula:

$$P_{i,j} = \frac{n_{i,j}/n_i}{N_j/N_t} \tag{1}$$

where $n_{i,j}$ is the number of residue i (i = 1–20) which occurs at a given position j of the π-turn, n_i is the total number of residue i in all the 640 proteins, N_j is the total number of residues in position j of the π-turn, N_t is the total number of residues in all the 640 proteins.

Support vector machine. We used SVM to distinguish π-turn residues from non-π-turn residues. This is a typical two-class classification problem. The SVM, as a binary-classification tool, is very suitable for solving such a problem. Because these data to be classified are typically not linearly separable, the SVM maps the input vectors into a higher dimensional feature space by a mapping function $\phi(x)$, and does a linear separation here. It tries to find the separating hyperplane with the largest distance between the two classes within the feature space by solving the following optimization problem:

$$\begin{aligned} &\min_{\omega,\xi_i} \quad \tfrac{1}{2}\omega^{\mathrm{T}}\omega + C\sum_{i=1}^{l}\xi_i \\ &\text{s.t.} \quad y_i(\omega\cdot x_i + b) \geqslant 1-\xi_i, \quad \xi_i \geqslant 0, \quad i=1,\ldots,l \end{aligned} \tag{2}$$

where x_i represents an input vector, $y_i = \pm 1$ according to whether x_i belongs to the π-turn class or non-π-turn class, l is the number of training data, ω is a weight vector, b is a bias, ξ_i denotes a slack variable, and C is a regularization parameter that controls the trade off between margin and classification error. $k(x_i,x_j) = \langle\phi(x_i),\phi(x_j)\rangle$ denotes the kernel function. Then the corresponding dual quadratic programming problem can be written as:

$$\begin{aligned} &\min_{\alpha_i} \quad \tfrac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j \alpha_i \alpha_j K(x_i\cdot x_j) - \sum_{j=1}^{l}\alpha_j \\ &\text{s.t.} \quad \textstyle\sum_{i=1}^{l} y_i\alpha_i = 0, \quad 0 \leqslant \alpha_i \leqslant C, \quad i=1,\ldots,l \end{aligned} \tag{3}$$

where α_i are the solutions of the dual formulation. An unlabeled residue x_j can subsequently be classified by the following discriminant function:

$$f(x_j) = \sum_{i=1}^{l}\alpha_i y_i K(x_i,x_j) + b. \tag{4}$$

The residue x_j is classified as the π-turn class if $f(x_j)$ is positive, non-π-turn class otherwise.

SVM was implemented with the SVM light package created by Joachims [31].

Encoding schemes. In the present study, we employed several different encoding schemes, i.e., single sequence, multiple sequence alignment in the form of PSI-BLAST position-specific profiles (PSSMs), single sequence plus predicted secondary structure by PSIPRED, multiple sequence alignment plus predicted secondary structure by PSIPRED, and multiple sequence alignment plus observed secondary structure assigned by DSSP [32].

The SVM, like other machine-learning methods, needs a fixed-length input vector. We adopted the classical local coding scheme of the protein sequence with a sliding window to produce the fixed-length input vectors. Each residue is encoded as a vector of 20 binary elements of a single one and 19 zeros, when using a single sequence as input. In order to allow a window to extend over the N-terminus and the C-terminus, a "null" residue represented by a 20-zero vector is used. The PSI-BLAST profiles have been used as input to the SVM classifier when using multiple sequence alignment information. The PSSMs were generated using PSI-BLAST with three rounds against the non-redundant protein sequence database (ftp://ftp.ncbi.nlm.nih.gov/blast/db/) with a cutoff E-value of 0.001. In the non-redundant database, low-complexity regions, coiled coil regions, and transmembrane spans were filtered with PFILT [29]. The profile matrix elements are scaled to 0–1 range by the standard logistic function:

$$f(x) = \frac{1}{1+\exp(-x)} \tag{5}$$

where x is the raw profile matrix value. Secondary structure information predicted by PSIPRED is encoded as follows: helix $\to (1,0,0)$, strand $\to (0,1,0)$, and coil $\to (0,0,1)$.

Reliability index to the prediction. The determination of the prediction reliability index (RI) is important when using machine-learning methods to identify π-turn residues in a protein sequence. One can easily identify the key regions with high prediction accuracy by means of RI. In this work, the reliability index is defined as that proposed by Hua and Sun [21]:

$$\mathrm{RI} = \begin{cases} 0 & \text{if} \quad \mathrm{abs}(D) < 0.2 \\ \mathrm{int}(\mathrm{abs}(D)/0.2) & \text{if} \quad 0.2 \leqslant \mathrm{abs}(D) < 1.8 \\ 9 & \text{if} \quad \mathrm{abs}(D) \geqslant 1.8. \end{cases} \tag{6}$$

Here, D is the distance of the sample to the optimal separating hyperplane (OSH). The distance is an effective measure of the prediction reliability. Fig. 1 shows the distribution of the prediction accuracy $Q_{\text{predicted}}$ and the prediction coverage Q_{observed} with different reliability indices. The "state-shifting" rule (see below) decreases the prediction accuracy $Q_{\text{predicted}}$ and increases the prediction coverage Q_{observed} at RI < 2. When RI $\geqslant$ 2, the higher the RI is, the more reliable the prediction is.

Filtering. The disparity of non-π-turn residues and π-turn residues will lead to a better prediction for the larger class non-π-turn residues for the purpose of maximizing the total accuracy. Therefore, we adopted a filtering scheme called the "state-shifting" rule first proposed by Shepherd et al. [4] to interpolate between the over- and underprediction of π-turns at the decision stage. We can change the state of residues whose prediction strength lies below a chosen threshold. The distance to the OSH is used to indicate the prediction strength. For example, one rule (represented as NT < 0.2) flips all non-turn predictions with abs(D) < 0.2 to turn.

Since π-turns are typically six residues long, but the prediction is performed for each residue separately without taking into account the correlations among residues, we applied the second filtering step similar to the "state-flipping" rule described by Shepherd et al. [4]. First, it flips all

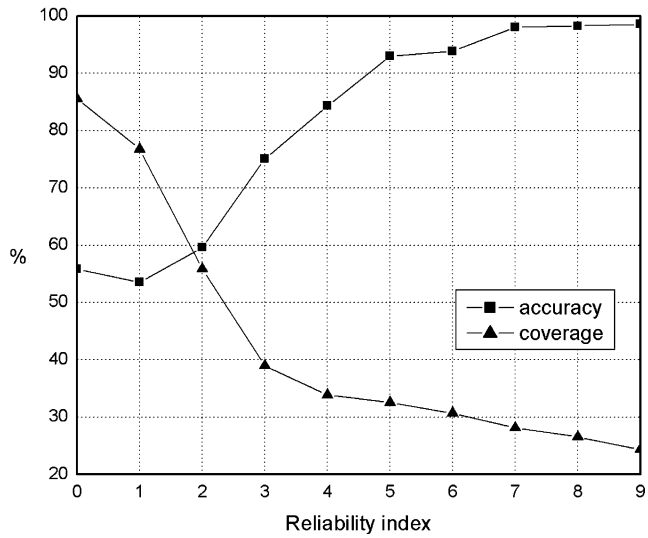

Fig. 1. Prediction accuracy $Q_{\text{predicted}}$ and prediction coverage Q_{observed} with different reliability indices. Results were generated with multiple alignments and predicted secondary structure.

isolated non-π-turn predicted residues to be π-turns. Second, it filters all the residues that are isolated π-turn predicted residues, isolated pairs of π-turn predicted residues, isolated three consecutive π-turn predicted residues, and isolated four consecutive π-turn predicted residues to be non-π-turn residues. Finally, for the isolated five consecutive π-turn predicted residues, the adjacent non-π-turn prediction with the lower RI is filtered to be π-turn.

Performance measures. The prediction performance was evaluated through a 7-fold cross-validation. The data set was randomly partitioned into seven sets with approximately equal size. A classifier was then trained on six sets and tested on the remaining set. This process was repeated for seven iterations, each time setting aside a different test set. The four most common measures to evaluate the turn/non-turn prediction problem are: (1) Q_{total} is the percentage of correctly predicted residues, which tends to give a highly misleading impression of the prediction quality when the data set is unbalanced, (2) $Q_{\text{predicted}}$ is the percentage of correctly predicted π-turns, which penalizes overpredictions, (3) Q_{observed} is the percentage of observed π-turns that are correctly predicted, which penalizes underpredictions, and (4) MCC is a more robust measure of the prediction quality, which takes into account both over- and underpredictions. They can be calculated by the following equations:

$$Q_{\text{total}} = \frac{p+n}{t} \times 100, \tag{7}$$

$$Q_{\text{predicted}} = \frac{p}{p+o} \times 100, \tag{8}$$

$$Q_{\text{observed}} = \frac{p}{p+u} \times 100, \tag{9}$$

$$\text{MCC} = \frac{pn - ou}{\sqrt{(p+o)(p+u)(n+o)(n+u)}}. \tag{10}$$

Here, p is the number of correctly classified π-turn residues, n is the number of correctly classified non-π-turn residues, o is the number of non-π-turn residues incorrectly classified as π-turn residues, u is the number of π-turn residues incorrectly classified as non-π-turn residues, and t is the total number of residues.

Another measure is also considered to compare the present method's performance with random predictions. The following score is to calculate the normalized percentage better than random:

$$S = \frac{(p+n) - R}{t - R} \times 100 \tag{11}$$

where

$$R = \frac{(p+o)(p+u) + (n+u)(n+o)}{t}$$

R reflects the total number of residues that are expected to be correctly predicted by a random prediction. For a perfect prediction, S equals 100%. For a prediction that is no better than random, S equals 0%.

Segment overlap measure (SOV). Because π-turns are segments of six residues, we adopted another measure SOV to complement the above five measures to assess the prediction quality. SOV is a measure for evaluation of secondary structure prediction methods by secondary structure segment rather than individual residues [33]. It is calculated as:

$$\text{SOV} = \frac{1}{N}\sum_{s} \frac{\min ov(s_1, s_2) + \delta(s_1, s_2)}{\max ov(s_1, s_2)} \times \text{len}(s_1) \tag{12}$$

where s_1 and s_2 are the observed and predicted π-turns; min $ov(s_1, s_2)$ is the length of actual overlap of s_1 and s_2; max $ov(s_1, s_2)$ is the length of the total extent for which either of the segments s_1 and s_2 has a residue in π-turn; len(s_1) is the number of residues in the segment of s_1; N is the number of residues in π-turn and sum is taken over all the pairs of overlapping segments; and $\delta(s_1, s_2)$ is defined as:

$$\begin{aligned}\delta(s_1, s_2) = \min((&\max ov(s_1, s_2) \\ &- \min ov(s_1, s_2)); \min ov(s_1, s_2); \text{int}(\text{len}(s_1)/2); \text{int}(\text{len}(s_2)/2)).\end{aligned} \tag{13}$$

Results and discussions

Positional potentials

A total of 1931 π-turns was extracted in our data set. This is more than four times the number of π-turns analyzed in the study of Rajashankar and Ramakumar [14]. Therefore, it is necessary and important to calculate the updated positional potentials and the overall turn potentials for each residue. The results are given in Table 1. Notable among the increased preferences compared with the earlier study [14] are Pro (increased by ∼26%), Ser (∼20%), and Asn (∼12%), and among the decreased preferences are His (decreased by ∼38%) and tryptophan (∼19%). There are considerable differences between the potentials of each position of π-turns. It can be noted that at positions i and $i+5$, hydrophobic residues are preferred. At position $i+1$, Lys is the most preferred amino acid, followed by Arg and Glu. At position $i+2$, Glu is the most preferred amino acid, followed by Lys and Gln. Position $i+3$ is dominated by His, and to a less extent, by Asn. There is a very strong preference for Gly at position $i+4$.

Parameter optimization of the prediction system

We selected the kernel function based on previous studies. The radial basis function (RBF) kernel performs better than other kernels in many studies such as protein secondary structure prediction [19], protein subcellular location prediction [21], and protein relative solvent accessibility prediction [25]. Accordingly, in this work, we use the RBF kernel:

$$k(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2). \tag{14}$$

Then the parameters to be ascertained are the kernel parameter γ and the regularization parameter C. Since

Table 1
Positional and overall potentials for π-turns

Amino acids	Positional						Overall		Difference %
	i	$i+1$	$i+2$	$i+3$	$i+4$	$i+5$	Present	Earlier[a]	
Ala	2.02	1.12	1.54	1.29	0.37	1.00	1.22	1.27	−3.43
Cys	2.05	0.92	0.43	0.83	0.73	1.12	1.01	0.91	9.69
Asp	0.87	0.80	1.00	0.60	0.77	0.52	0.76	0.74	2.06
Glu	0.44	1.42	2.10	0.89	0.57	0.68	1.02	1.06	−3.93
Phe	1.22	0.53	0.41	0.91	0.54	1.31	0.82	0.88	−7.42
Gly	0.47	0.39	0.56	0.55	6.34	1.45	1.63	1.51	7.22
His	0.85	1.32	0.60	1.64	0.89	0.79	1.02	1.41	−38.44
Ile	0.96	1.22	0.43	0.57	0.29	1.55	0.84	0.85	−2.36
Lys	0.52	1.67	1.81	1.21	1.07	1.15	1.24	1.28	−3.45
Leu	2.14	1.17	0.73	1.47	0.43	1.50	1.24	1.21	1.92
Met	1.82	0.78	1.26	1.31	0.34	1.19	1.12	1.23	−10.22
Asn	0.79	0.78	1.14	1.54	1.92	0.73	1.15	1.02	11.64
Pro	0.21	0.60	0.31	0.18	0.12	0.07	0.25	0.18	25.75
Gln	0.70	1.28	1.80	1.20	0.74	0.75	1.08	1.15	−6.52
Arg	0.65	1.60	1.43	1.20	0.83	0.84	1.09	1.12	−2.81
Ser	0.97	0.76	1.23	1.11	0.38	0.59	0.84	0.68	19.52
Thr	0.48	0.77	0.67	1.11	0.30	0.70	0.67	0.60	10.06
Val	0.74	0.96	0.48	0.71	0.31	1.35	0.76	0.78	−3.47
Trp	1.73	0.90	0.50	0.53	0.43	0.90	0.83	0.99	−19.03
Tyr	1.04	0.79	0.62	1.17	0.39	1.28	0.88	0.90	−1.59

[a] Earlier refers to the study by Rajashankar and Ramakumar [14].

the ratio of non-π-turn residues to π-turn residues in the data set is 9:1, another parameter cost factor j is adopted to interpolate between the over- and underprediction of π-turns. Then the penalty term $C\sum_i^{\xi_i}$ in Eq. (2) becomes: $C_+\sum_{i:y_i=1}\xi_i + C_-\sum_{i:y_i=-1}\xi_i, (C_+ = jC_-)$. The cost factor j is usually set to [34]:

$$j = \frac{N_-}{N_+}. \tag{15}$$

In this study, N_+ is the number of π-turn residues and N_- is the number of non-π-turn residues. The detail of the selection process of γ and C can be seen in the supplementary material. The optimized parameters are: $\gamma = 0.0625$, $C = 5$, and $j = 9$.

Window size optimization

A proper window size can lead to a good prediction performance. A too short window size can lose some important classification information. But there is an up limit for improving prediction accuracy by increasing the window lengths because a too long window size will decrease the signal-noise ratio. The optimal window size is obtained by testing different window length of 7–13. In Table 2, testing results based on different window lengths are shown. As can be seen from Table 2, the best prediction performance is achieved when the window length = 11. We adopted the optimum window size of 11 in the following analysis of this study.

Handling unbalanced data set

Unbalanced data set presents particular difficulties for training a binary classifier. There are two strategies to handle this problem. The first strategy is to balance the training data set by re-sampling techniques, either by discarding some training samples of the larger size class or duplicating some training samples of the smaller class. But when presented with the data set of maintaining the naturally occurring proportions, the SVM classifier will tend to underpredict the larger class of non-π-turns. The second strategy is to adopt a two-stage method as described by Wang et al. [13]. The first step is to interpolate between the over- and underprediction of π-turns at the training stage by setting different regularization parameters C_+ and C_-. The second step is realized by the use of the state-shifting rule (see above) at the decision stage.

The effect of filtering the prediction using different state-shifting rules is shown in Figs. 2 and 3. As can be seen from

Table 2
Dependence of testing accuracy on the window length

	$l = 7$	$l = 9$	$l = 11$	$l = 13$
Q_{total}	92.8 ± 0.2	92.9 ± 0.2	93.1 ± 0.2	92.8 ± 0.2
$Q_{\text{predicted}}$	65.5 ± 2.5	65.4 ± 2.8	68.7 ± 2.7	66.1 ± 2.6
Q_{observed}	51.5 ± 1.6	53.1 ± 1.9	50.9 ± 1.9	49.1 ± 1.5
MCC	0.542 ± 0.019	0.551 ± 0.021	0.556 ± 0.021	0.532 ± 0.020

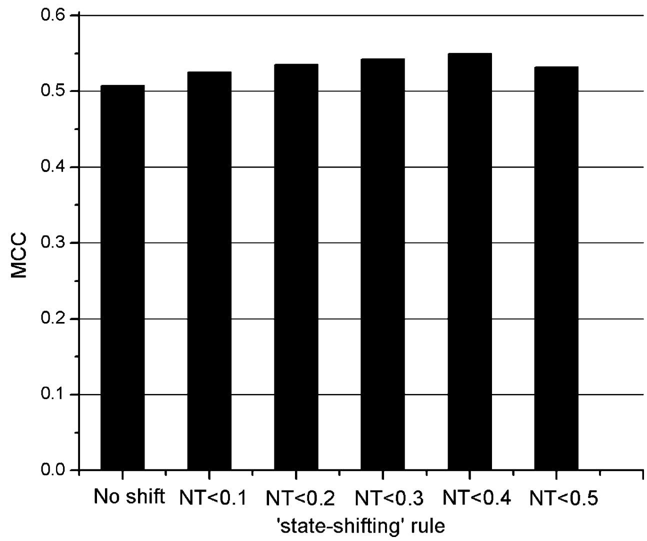

Fig. 2. The effect of "state-shifting" rule on MCC. Results were generated with multiple alignments and predicted secondary structure.

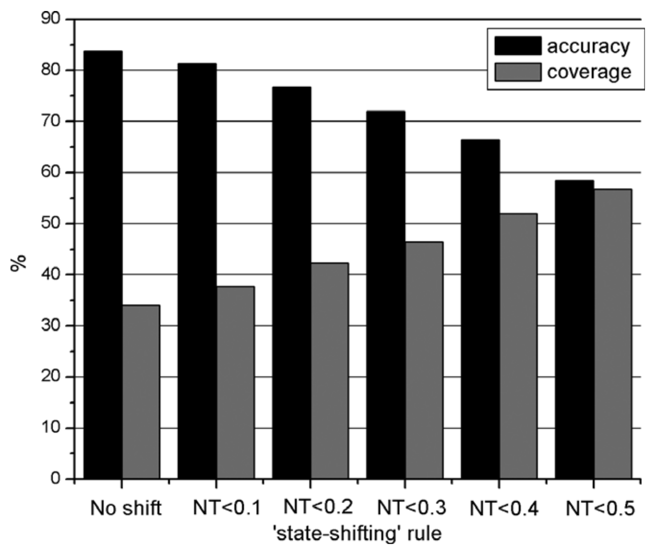

Fig. 3. The effect of "state-shifting" rule on $Q_{predicted}$ and $Q_{observed}$. Results were generated with multiple alignments and predicted secondary structure.

Fig. 2, the largest improvement of MCC is attained at NT < 0.4. Fig. 3 shows the effect of the state-shifting rule in equalizing the rates of over- and underprediction.

Influence of PSSMs and secondary structures on prediction performance

In Table 3, we list the prediction results for the SVM method with different encoding schemes. A great improvement in the prediction performance has been achieved by using the multiple sequence alignment information. The effect is particularly high when no predicted secondary structure information is utilized as the MCC improved from 0.274 to 0.532 and the Q_{total} from 84.4 to 92.8%. The positive effect of PSSMs has also been shown in other tight turn predictions. This is due to the fact that PSSMs contain the information whether a residue is conserved or occurs by chance.

Use of predicted secondary structure information further improves the prediction quality. The influence is higher when no multiple sequence alignment information is used as the MCC improved from 0.274 to 0.341. With multiple sequence alignment and predicted secondary structure as the input vector, the final SVM classifier achievs MCC of 0.556, the segment measure SOV of 51.8%, and *S* of 54.6%. Use of observed secondary structure information brings the MCC to 0.571. The observed secondary structure assignments were made using the DSSP program.

Our data set contains homologues of some of the protein chains used to train PSIPRED. With our data set, PSIPRED reached a three-state overall per-residue accuracy of 80.3%, MCCs of 0.77 (helix), 0.68 (strand), and 0.63 (coil). The corresponding results when applied to a non-homologous protein data set (to 2006_04_05) were 77.8%, 0.71%, 0.64%, and 0.59% (http://cubic.bioc.columbia.edu/eva/sec/method/psipred.html), respectively. As a consequence, the impact of our data set containing homologues of some of the protein chains used to train PSIPRED on our results is small.

The good performance of the present method can be ascribed as the following three aspects. First, the powerful binary-classification tool SVM was adopted. The superior features of SVM are that it can not only effectively avoid overfitting with the use of structural risk minimization but also condense the information of training sets by identifying support vectors. Second, a two-step method to handle the highly unbalanced data set was used. As shown in the 'handling unbalanced data set' section, use of the state-shifting rule further improves the prediction quality. Third, a combined feature including multiple alignments and secondary structure was used. Both PSSMs and the predicted secondary structure information contribute the prediction performance.

Because this is the first attempt to predict π-turns in a protein sequence, there are no values with which to compare the results obtained in this work. But considering the similarity of β-turn prediction studies and this study and β-turn being the best-predicted tight turn, we compared the results obtained from this study with methods of β-turn prediction. The overall results with single sequence are poorer than β-turn prediction methods BetaTPred2 [6] and BetaTurn [8]. The main reason is that the data set used in this study is more unbalanced than the data sets used in the β-turn prediction method. But surprisingly, the overall results with multiple alignments are much better than BetaTPred2 [6] and BetaTurn [8]. In other words, multiple sequence alignment contributes more to the π-turn prediction than to the β-turn prediction. We think this is mainly because of the different secondary composition of π- and β-turn residues. π-Turn has 47% residues in helical regions

Table 3
Results with different encoding schemes

Method	Q_{total}	$Q_{predicted}$	$Q_{observed}$	MCC	S
(1)	84.4 ± 0.1	29.2 ± 0.8	44.2 ± 1.1	0.274 ± 0.012	0.267 ± 0.011
(2)	86.9 ± 0.1	36.3 ± 1.4	46.7 ± 1.3	0.341 ± 0.014	0.338 ± 0.013
(3)	92.8 ± 0.2	65.8 ± 2.8	49.3 ± 1.5	0.532 ± 0.020	0.526 ± 0.021
(4)	93.1 ± 0.2	68.7 ± 2.7	50.9 ± 1.9	0.556 ± 0.021	0.546 ± 0.020
(5)	93.3 ± 0.2	69.2 ± 2.9	53.1 ± 1.8	0.571 ± 0.022	0.564 ± 0.022

Results were based on the following encoding schemes: (1) single sequence, (2) single sequence plus predicted secondary structure, (3) multiple sequence alignment profiles, (4) multiple sequence alignment profiles plus predicted secondary structure, and (5) multiple sequence alignment profiles plus observed secondary structure.

```
PiTURN PREDICTION RESULT:

SEQUENCE:  MQDYTVHIVDDEEPVRKSLAFMLTMNGFAVKMHQSAEAFLAFAPDVRNGV
 PREDICT:  ----------------------pppppp----------------------
  RINDEX:  65565598756688999998565555347666666789898466766995

SEQUENCE:  LVTDLRMPDMSGVELLRNLGDLKINIPSIVITGHGDVPMAVEAMKAGAVD
 PREDICT:  ------------------------------------------pppppp--
  RINDEX:  67786543323567899733322386998997667566779703334234

SEQUENCE:  FIEKPFEDTVIIEAIERASEHLV
 PREDICT:  -----------------------
  RINDEX:  66765578888997978543335
```

Fig. 4. Sample π-turn/non-π-turn predictions. Predictions are for chain 1dbwA(123 residues) using multiple alignments and predicted secondary structure information. The predictions are grouped in blocks of 50 residues. For each block: row1 is the amino acid sequence; row2 is the π-turn/non-π-turn predictions (p, π-turn prediction; '-', non-π-turn prediction); and row3 is the reliability index for each π-turn/non-π-turn prediction.

(see supplementary material), while most β-turn residues are in coil regions.

π-Turn web server

A web server named PiTurn is available at http://210.42.106.80/piturn/. The program enables users to predict π-turns from the amino acid sequence. The input is a one-letter code amino acid sequence in plain text. The output gives the predicted π-turn or non-π-turn residues and the corresponding reliability index. A prediction result sample has been given in Fig. 4.

Conclusions

We described a reliable method for the prediction of π-turns from the amino acid sequence using the SVM. This is the first method designed specially for this task. The present method achieved the final MCC of 0.556 when using PSSMs and predicted secondary structure as input. We also noticed that multiple sequence alignment contributes more to the π-turn prediction than to the β-turn prediction. As a result, the accuracy level achieved for π-turns is better than that of β-turns, although the data set used in this work is more unbalanced than the data sets used in the β-turn prediction method. In addition, our method can be complementary to other protein secondary structure prediction methods. The newly computed positional potentials can be applied in modeling and design of π-turns in proteins.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.bbrc.2006.06.066.

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